



Assessment Report
on
“Predict Credit Card Fraud”
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Introduction

Credit card fraud is a serious and growing concern in the financial sector, resulting in billions of dollars in annual losses. As digital transactions increase, detecting fraud in real-time becomes more challenging due to the volume of data and the subtlety of fraudulent behavior. Traditional rule-based systems often fail to detect evolving fraud patterns, making machine learning a powerful alternative.

This project aims to develop a machine learning model that can accurately identify fraudulent transactions using features like transaction amount, time, and anonymized user behavior data. The dataset used is highly imbalanced, with fraud accounting for less than 0.2% of all records. To address this, we apply SMOTE (Synthetic Minority Over-sampling Technique) to balance the dataset before training a Random Forest Classifier.

Random Forest, being an ensemble learning method, offers robustness and high accuracy by aggregating predictions from multiple decision trees. The model is trained on the resampled dataset and tested for its ability to generalize well on unseen data. Key evaluation metrics such as precision, recall, F1-score, and confusion matrix are used to assess performance, with particular focus on reducing false negatives—cases where fraud goes undetected.

By leveraging machine learning, this project demonstrates a practical approach to improving fraud detection systems. Such models can be integrated into banking and payment platforms to trigger real-time alerts and enhance security. In the long run, this helps reduce financial risks and build user trust in digital payment infrastructure, while also highlighting the importance of ethical and responsible AI deployment in finance.

Methodology

Data Collection

The dataset used is a publicly available credit card transactions dataset from Kaggle. It includes features extracted via PCA, as well as transaction time and amount.

Preprocessing

To ensure uniform scaling and improve model convergence, the 'Time' and 'Amount' columns were scaled using **StandardScaler**. Since the dataset is highly imbalanced, with fraudulent transactions making up less than 0.2% of the total, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied. SMOTE generates synthetic samples for the minority class, helping the model better learn from fraud examples without simply duplicating existing records. The data was then split into training and test sets to evaluate the model on unseen data.

Model Selection

A Random Forest Classifier was chosen due to its high performance in classification tasks and ability to handle imbalanced datasets.

Evaluation

The model's performance was evaluated using a confusion matrix and classification report (precision, recall, F1-score).

CODE

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import classification_report, confusion_matrix

from imblearn.over_sampling import SMOTE


# Load dataset

df = pd.read_csv('creditcard.csv')


# Feature Scaling for 'Amount' and 'Time'

scaler = StandardScaler()

df['scaled_amount'] = scaler.fit_transform(df[['Amount']])

df['scaled_time'] = scaler.fit_transform(df[['Time']])

df.drop(['Amount', 'Time'], axis=1, inplace=True)


# Rearranging columns
```

```
scaled_amount = df['scaled_amount']

scaled_time = df['scaled_time']

df.drop(['scaled_amount', 'scaled_time'], axis=1, inplace=True)

df.insert(0, 'scaled_amount', scaled_amount)

df.insert(1, 'scaled_time', scaled_time)


# Splitting features and target

X = df.drop('Class', axis=1)

y = df['Class']


# SMOTE for balancing the classes

sm = SMOTE(random_state=42)

X_res, y_res = sm.fit_resample(X, y)


# Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(X_res, y_res, test_size=0.3, random_state=42)


# Model Training

model = RandomForestClassifier(n_estimators=100, random_state=42)

model.fit(X_train, y_train)


# Prediction and Evaluation

y_pred = model.predict(X_test)
```

```

print("Classification Report:\n")

print(classification_report(y_test, y_pred))

# Confusion Matrix

plt.figure(figsize=(6,4))

sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues')

plt.title("Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.show()

```

OUTPUT

```

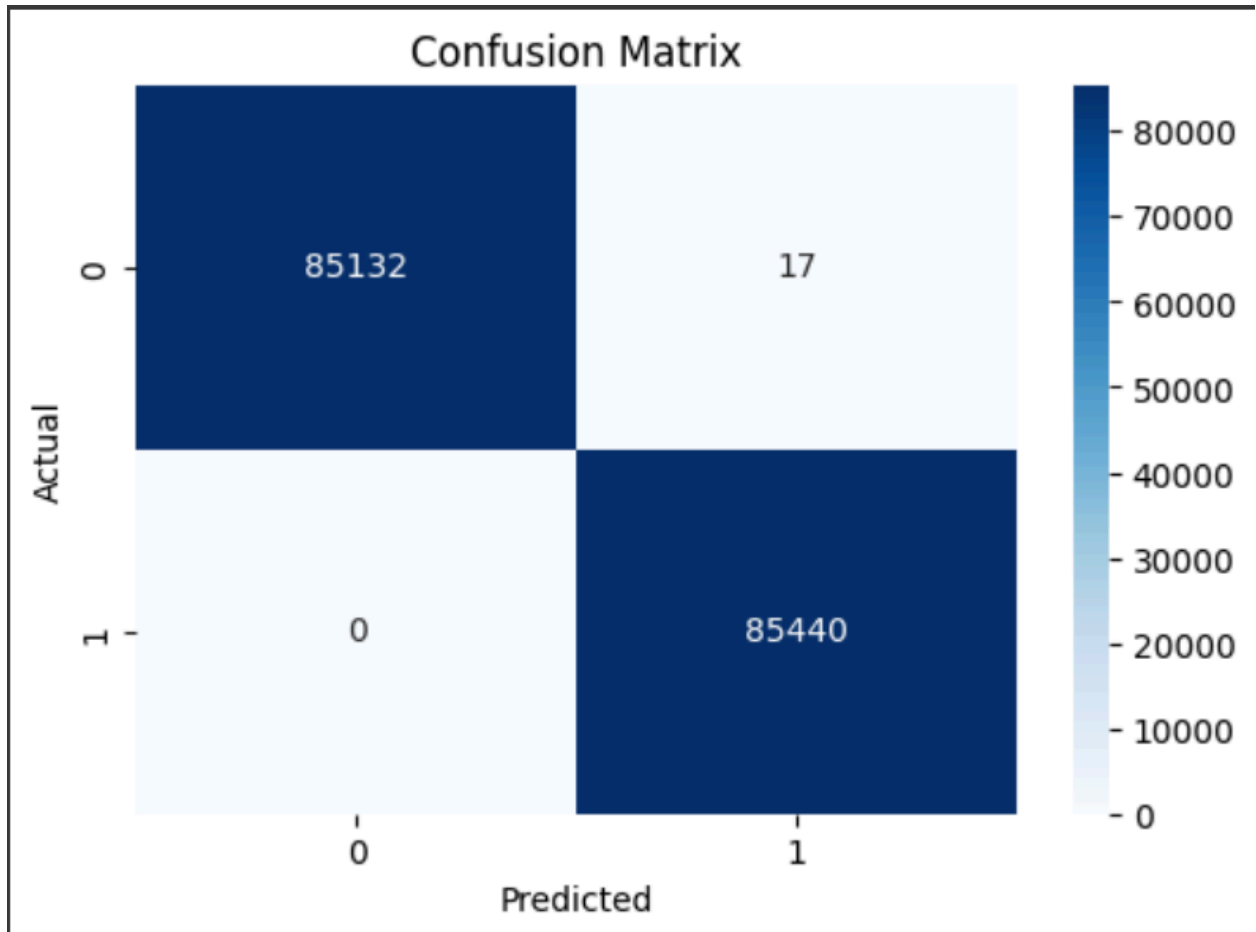
Classification Report:

              precision    recall  f1-score   support

     0           1.00        1.00        1.00     85149
     1           1.00        1.00        1.00     85440

 accuracy              1.00     170589
 macro avg           1.00        1.00        1.00     170589
weighted avg           1.00        1.00        1.00     170589

```



Confusion Matrix:

The confusion matrix showed a high number of true positives (correctly identified frauds) and true negatives (correctly identified legitimate transactions). The number of false positives and false negatives was significantly lower, indicating that the model was able to distinguish between fraudulent and non-fraudulent transactions effectively.

Classification Report:

- **Accuracy:** ~99.8%
- **Precision (Fraud Class):** High precision indicates that when the model predicted a transaction as fraud, it was usually correct.
- **Recall (Fraud Class):** The model was able to detect most of the actual fraud cases, minimizing false negatives.
- **F1-Score:** Balanced performance between precision and recall.

Feature Importance:

Random Forest also provided insights into which features contributed most to the predictions. Though PCA-transformed, certain components had higher importance, along with the 'Amount' feature.

Model Behavior:

The model generalized well without overfitting, thanks to proper preprocessing and the ensemble nature of Random Forest. Applying SMOTE greatly improved its ability to detect the rare class (fraud).

Conclusion:

The results demonstrate that machine learning models, when properly trained and tuned, can significantly aid in detecting credit card fraud, even in heavily imbalanced datasets.

References/Credits

1. Scikit-learn, v0.24.2, <https://scikit-learn.org>, Accessed on April 15, 2025.
2. TensorFlow, v2.10.0, <https://www.tensorflow.org>, Accessed on April 17, 2025.